AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 5

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Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Methodology / Proposed Method and Experimental Setup

3.1 System Overview

WaterTriage is implemented as a four-stage pipeline: data acquisition, storage, enrichment, and serving. In the acquisition stage, water quality records are pulled from public government sources and cached locally. The storage stage persists these records in a normalized PostgreSQL schema. The enrichment stage computes per-record contamination severity scores and recurrence classifications. The serving stage exposes the enriched data through a REST API and a lightweight browser-based dashboard for demonstration and interactive exploration. This staged design keeps acquisition, scoring logic, and presentation independently testable and replaceable, which was a deliberate choice given the uncertainty around final data source availability at the start of the project.

3.2 Database Schema

The system persists data in eight PostgreSQL tables, summarized below. The geographic hierarchy (states, districts, villages) allows records to be aggregated and filtered at any administrative level, while the water_samples and readings tables separate a lab report from its individual parameter measurements, permitting a single sample to carry multiple readings without denormalization.

Table	Purpose
states, districts, villages	Geographic hierarchy; village records carry block and panchayat identifiers
bis_parameters	BIS IS 10500 acceptable/permissible limits, health-severity weight, and evaluation strategy (threshold / range / microbial), seeded at application startup
water_samples	One laboratory report per record (unique WQMIS reference, source, sample date)
readings	Individual parameter values linked to a sample
risk_scores	Computed composite score, severity band, and worst-parameter flag per sample (recomputable)
historical_contamination	Registry of quality-affected habitation events by village, parameter, and year (2009–2012)
interventions	Planned remediation action log (schema implemented; interface pending)

State names from different source datasets were found to use inconsistent casing and formatting (for example, "UTTAR PRADESH" versus "Uttar Pradesh") during initial integration; both loaders now canonicalize state names to Title Case before insertion to prevent duplicate geographic entities.

3.3 Contamination Severity Scoring

Each parameter reading is converted to a severity value between 0 and 1 by linear interpolation between the BIS IS 10500 acceptable limit and permissible limit for that parameter, consistent with the weighted-scoring approach described in the

reviewed WQI literature (Section 2.3). Parameters for which BIS specifies no permissible relaxation (for example, nitrate and iron) and microbial parameters saturate directly to a severity of 1.0 upon any exceedance, reflecting their comparatively immediate health risk. The composite score for a sample is computed as the weighted mean of severity across all tested parameters, with weights assigned according to documented health severity rather than treated as uniform. To correct a known weakness of simple averaging — where a single severe exceedance can be diluted by several unaffected parameters — a band-escalation rule is applied on top of the composite score: any parameter reaching full severity forces a minimum "High" classification, and acute-hazard parameters (E. coli, total coliform, arsenic) force a "Critical" classification irrespective of the numeric composite score.

3.4 Recurrence Detection

To distinguish a persistent contamination problem from an isolated anomalous reading, each village-parameter combination is evaluated against the historical contamination registry (2009–2012) alongside current sample data. A recurrence score blends the number of years a village-parameter combination appears in the historical registry (capped at four years) with the current exceedance status, producing one of four verdicts: chronic-hotspot (persistent across both historical and current data), recurring (present in multiple historical years), watchlist (a new current exceedance with no historical record), or no-history.

3.5 Data Acquisition Strategy

Three acquisition channels were evaluated. The primary channel is a crawler against WQMIS's public, unauthenticated report-printing endpoint, which accepts a report identifier encoded via a client-side parameter scheme; report identifiers were found to cluster geographically, allowing dense-range crawling once a district's identifier range is located. The crawler operates single-threaded at approximately one request per second, caches every retrieved report locally, and maintains a resume checkpoint so that interrupted runs do not require re-fetching completed ranges. A parameter-wise WQMIS API was also identified but was found to require an authenticated departmental role rather than public access, and was therefore set aside as a possible future channel rather than a current one. The AIKosh open data portal remains a pending, access-gated source; its schema has been pre-mapped so that it can be integrated as a drop-in supplementary source if and when access is granted. As a supplementary source, a historical Quality-Affected Habitations registry (2009–2012) and publicly available groundwater quality datasets covering Uttar Pradesh and Bihar were loaded directly, providing the temporal depth needed for the recurrence detection module described in Section 3.4.

3.6 Implementation and Experimental Setup

The backend is implemented in Python using FastAPI for the REST API layer and SQLAlchemy for object-relational mapping against PostgreSQL 16, run via Docker Compose for local development. Nine REST endpoints expose ranked intervention priority lists, individual samples, district-level summaries, recurrence-based hotspot rankings, and state-to-state comparison. A zero-dependency, single-file browser dashboard consumes these endpoints to present three views — a ranked priority queue, a state comparison view, and a district-level choropleth map for Uttar Pradesh and Bihar — with a drill-down panel showing individual parameter readings against BIS limits. The dashboard is designed to load an offline baked data snapshot by default and to upgrade to live API data automatically when the backend is reachable, so that the interface remains usable for demonstration purposes independent of backend availability.

3.7 Testing and Validation

The implementation is validated using twenty-six automated test cases covering four areas: correctness of the scoring engine in isolation (interpolation behaviour, band-escalation rules, and edge cases such as missing parameters); API contract tests using FastAPI's TestClient against a seeded SQLite database; parser tests against WQMIS report fixtures, including malformed or incomplete numeric fields; and recurrence-logic unit tests together with a PostgreSQL integration suite that skips cleanly in environments without a database connection. A performance issue was identified during testing in which the recurrence ranking query executed one database round-trip per village (an N+1 query pattern), resulting in an execution time of approximately 37 seconds across the full village set; this was resolved by rewriting the operation as four set-based queries with in-memory aggregation, reducing execution time to approximately 0.58 seconds while producing identical output, verified by regression tests.